

UMC 110nm Logic and Mixed-Mode AI Advanced Enhancement 3.3V NCAP SPICE Model Document

**Version 0.3 Phase 1
2011/09/27**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.1_p1	2010/01/28	Ho Cheng Hu	1. Original release 2. New DSM version definition from 2009/11/30.
0.2_p1	2010/06/14	Ho Cheng Hu	Modify model card to fix up-trend issue when $L > 2\mu\text{m}$.
0.3_p1	2011/09/27	Shih-Hsin Yeh	Refine C-W and C-L trends.

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor AI Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.9
Cadence SPECTRE version 6.2.1.299
Mentor Graphics ELDO version v6.10_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

TYP: Typical NCAP case
MAX: Maximum NCAP case
MIN: Minimum NCAP case

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib" CaseName
, where CaseName could be TYP, MAX, or MIN.
- Define device condition by 'Xxx' statement.

Example:

X1 VGG GND DeviceName LF=2E-6 WF=5E-6 NF=8

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.

- Add the following statement in the input file to the simulator.

include "./ModelFileName.lib.scs" section=CaseName

, where CaseName could be TYP, MAX, or MIN.

- Define device condition by 'Xxx' statement.

Example:

X1 (VGG GND) DeviceName LF=2E-6 WF=5E-6 NF=8

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.

- Add the following statement in the input file to the simulator.

.lib "./ModelFileName.lib.eldo" CaseName

, where CaseName could be TYP, MAX, or MIN.

- Define device condition by 'Xxx' statement.

Example:

X1 VGG GND DeviceName LF=2E-6 WF=5E-6 NF=8

ModelFileName: l110ae_ncap33_v031

DeviceName: ncap_33

Geometry description (Before shrink):

$0.35\ \mu\text{m} \leq \text{LF} \leq 2\ \mu\text{m}$

$2.5\ \mu\text{m} \leq \text{WF} \leq 10\ \mu\text{m}$

$2 \leq \text{NF} \leq 8$

Voltage Range:

$-3.3\text{V} \leq \text{VGS} \leq 3.3\text{V}(*1.1)$

Layout example:

The following figure is the layout example of the NCAP. LF and WF are the single poly finger length and width, respectively, and Nf is the total finger number.

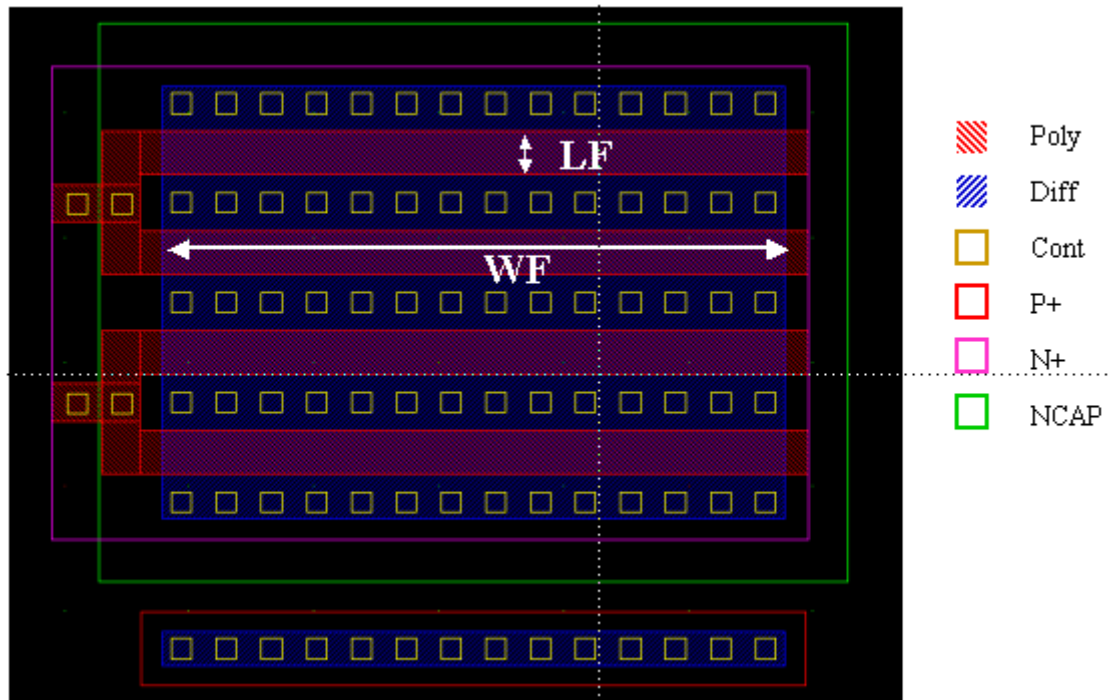


Figure 1-1 NCAP layout

2 Capacitance Model

2.1 Silicon Wafer for Modeling

The information of the hardware used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NCAP

	Wafer for NCAPs measurements data
Mask ID	BSW09A
Lot ID	D4QAC
Wafer Number	#5

2.1.1 Test devices

The following geometries were available for parameters extraction.

Table 2-2 Geometry of NCAP devices measured for parameters extraction.

W/Lf	2	1	0.75	0.5	0.35
10	X				
5	X	X	X	X	X
2.5	X				

2.1.2 Measurement conditions

The capacitance was measured under the conditions summarized in the table below.

Table 2-3 Test conditions for Cg curves of NCAP

$V_{DD} = V$	C_G vs. V_G
V_G	From -3.65 to 3.65 V in 0.05 V step
V_B	0V
Temperature [°C]	-50, 25, 125

2.2 Electrical Parameters

The following table shows the measured and simulated Cmax (fF/um²), where Cmax is the capacitance at VG=3.3V, VB=0 V, T=25°C.

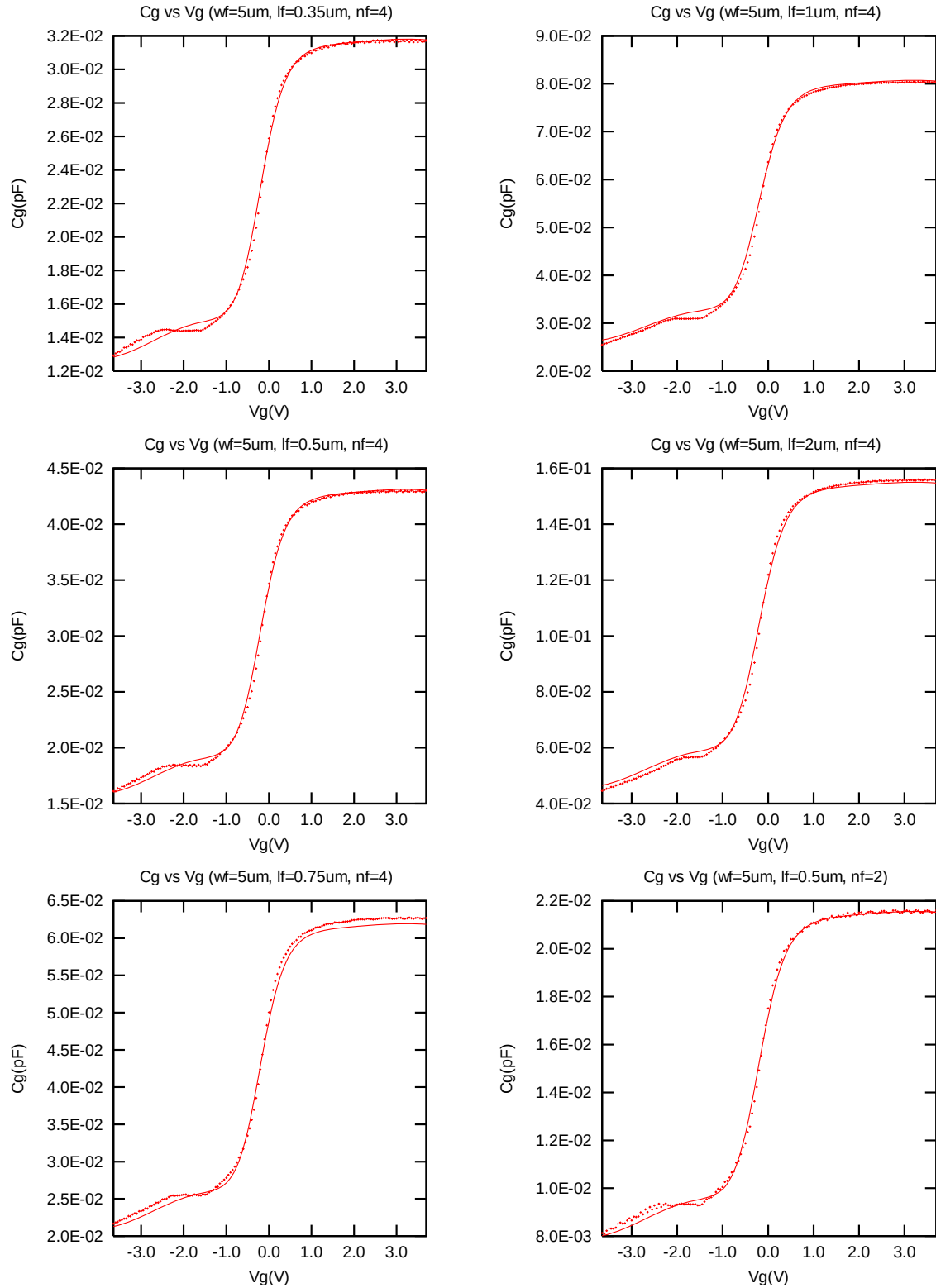
Table 2-4 Electrical Parameters

WF*0.9 (um)	LF*0.9+0.012 (um)	NF	M	Measured Cmax* (fF/um ²)	Model Cmax* (fF/um ²)
4.5	0.327	4	128	4.519	4.544
4.5	0.462	4	104	4.292	4.313
4.5	0.687	4	72	4.184	4.130
4.5	0.912	4	120	4.020	4.036
4.5	1.812	4	48	3.899	3.877
4.5	0.462	2	96	4.317	4.313
2.25	1.812	4	64	3.956	4.010
4.5	0.462	8	80	4.283	4.313
9	1.812	4	32	3.874	3.878

*Cmax is normalized by the area (WF*LF) before 90% shrinkage.

2.2.1 Cg vs. Vg plots

The following figures show the measured and simulated Cg (pF) vs. Vg (V). The measurement data is dots and simulation is line, respectively.



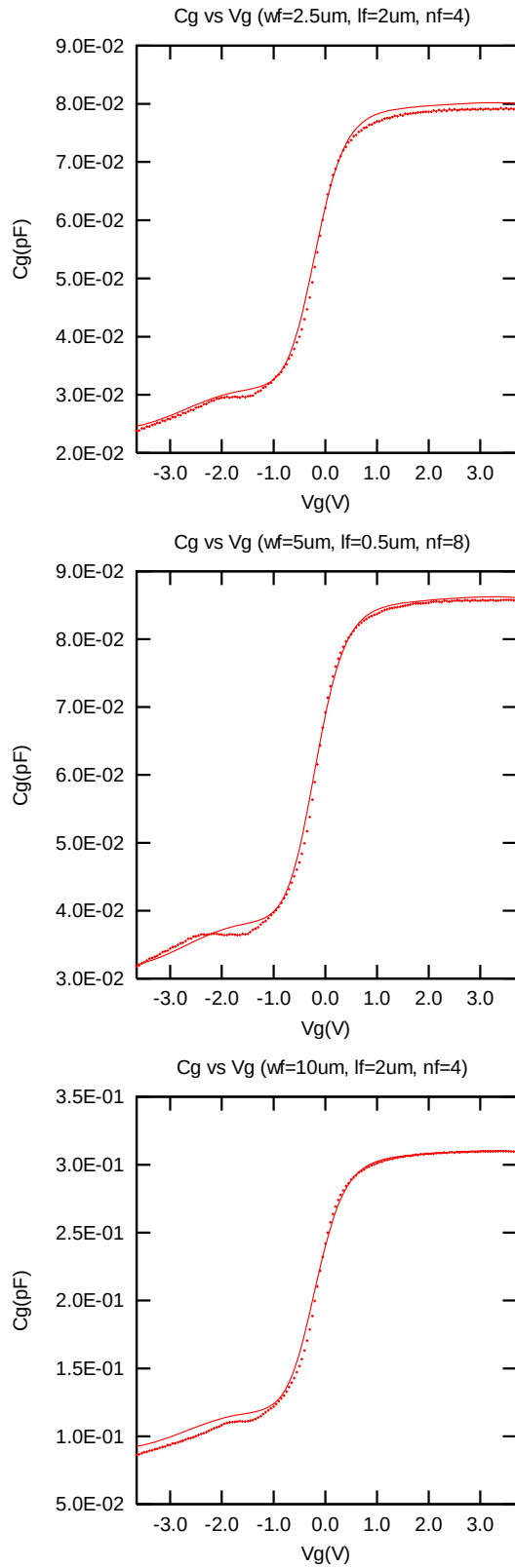


Figure 2-1 C_g vs. V_g at 25 °C

3 Appendix

3.1 HSPICE Warning Message

Item	Warning Message	Explanation
1.	None.	None.

3.2 SPECTRE Warning Message

Item	Warning Message	Explanation
1.	None.	None.

3.3 ELDO Warning Message

Item	Warning Message	Explanation
1.	None.	None.